We move now to the case $\alpha > 2$. To start off, we recall some properties of the optimal rotations. We focus on the set $\mathcal{M}$ but all the arguments can be replicated for $\mathcal{M}_h$. Firstly, observe that given a rotation $R \in \mathrm{SO}(3)$ and a skew-symmetric matrix $W \in \mathbb{R}^{3\times 3}_{\mathrm{skew}}$ the map $\phi(t) = Re^{tW}U_j$ takes values in K for any choice of $j \in \{1, \dots, l\}$. In particular, if $RU_j \in \mathcal{M}$, then by differentiating the map $t \mapsto F(\phi(t))$ at $t = 0$ we obtain

$$F(RWU_j) = 0 \quad \text{and} \quad F(RW^2U_j) \leq 0 \quad \forall\, W \in \mathbb{R}^{3\times 3}_{\mathrm{skew}}\,, \forall\, j \in \Lambda\,, \forall\, R \in \mathcal{R}^j\,.$$

The missing information that we need to prove Theorem 1.7 is the well-posedness of the projection operator P^j_h along a sequence of rotations converging to some element of $\mathcal{R}^j$. In order to do so we start proving some useful lemmas. The proof of the first one is trivial and is omitted.

Lemma 5.2. *The sequence of functional* $-\dfrac{1}{h^{\gamma+1}}F_h$ Γ*-converges to* $-F$*. In particular, given a sequence* $(R_hU_{k_h})$ *such that* $R_hU_{k_h} \in \mathcal{M}_h$ *for every* h *and* $R_hU_{k_h} \to RU_j$ *for some* $j \in \{1, \dots, l\}$*, we have* $j \in \Lambda$ *and* $RU_j \in \mathcal{M}$*.*

Lemma 5.3. *Let* $j \in \{1, \dots, l\}$*. Let* $(R_h) \subset \mathrm{SO}(3)$ *such that* $R_h \in \mathcal{R}^j_h$ *for every* h *and let* $(W_h) \subset \mathbb{R}^{3\times 3}_{\mathrm{skew}}$ *be a sequence such that* R_hW_h *is normal to* $\mathcal{R}^j_h$ *at the point* R_h *and* $|W_h| = 1$ *for every* h*. Then, up to a subsequence, we have* $R_hW_h \to RW$*, where* $R \in \mathcal{R}^j$*,* $W \in \mathbb{R}^{3\times 3}_{\mathrm{skew}}$*,* $|W| = 1$*, and* $RW \in N\mathcal{R}^j_R$*.*

Proof. Up to subsequences, we have that $R_h \to R$ and $W_h \to W$ with $R \in \mathrm{SO}(3)$, $W \in \mathbb{R}^{3\times 3}_{\mathrm{skew}}$, and $|W| = 1$. By Lemma 5.2 we have that $R \in \mathcal{R}^j$, thus we just need to prove that $RW \in N\mathcal{R}^j_R$. Let $m = \dim \mathcal{R}^j$. By (F2), $m = \dim \mathcal{R}^j_h$ for $h \ll 1$. Consider an orthonormal basis of the tangent space $T\mathcal{R}^j_{hR_h}$ given by $\{R_hW^1_h, \dots, R_hW^m_h\}$. Then, up to a subsequence, we have $R_hW^i_h \to RW^i$ for some $W^i \in \mathbb{R}^{3\times 3}_{\mathrm{skew}}$. Clearly the matrices $\{RW^1, \dots, RW^m\}$ are orthonormal. Moreover, since

$$0 = \frac{1}{h^{\gamma+1}}F_h(R_h(W^i_h)^2U_j) \to F(R(W^i)^2U_j) \quad \forall\, i = 1, \dots, m$$

it follows that $\{RW^1U_j, \dots, RW^mU_j\}$ is an orthonormal basis of $T\mathcal{R}^j_R$.

Consider now a matrix $M \in \mathbb{R}^{3\times 3}_{\mathrm{skew}}$ such that $RM \in T\mathcal{R}^j_R$. Then, we can write $RM = \sum_{i=1}^m \lambda_i RW^i$ for some $\lambda_1, \dots, \lambda_m \in \mathbb{R}$. Define $M_h = \sum_{i=1}^m \lambda_i W^i_h$. By construction, we have $R_hM_h \to RM$. Moreover, R_hM_h is tangent to $\mathcal{R}^j_h$ at the point R_h, so that

$$0 = R_hW_h : R_hM_h \to RW : RM\,.$$

Since M is arbitrary, this concludes the proof. ∎

Remark 5.4. Lemma 5.3 proves also that a sequence of tangent matrices to $\mathcal{R}^j_h$ at a point R_h converges to a tangent matrix to $\mathcal{R}^j$ at the point R, where R is the limit of R_h.

Proposition 5.5. *For* $j \in \{1, \dots, l\}$ *let*

$$\tilde{\mathcal{R}}^j = \{R \in \mathcal{R}^j : \exists\, (R_h) \subset \mathrm{SO}(3) \text{ s.t. } R_h \in \mathcal{R}^j_h \text{ for every } h > 0 \text{ and } R_h \to R\}\,.$$

Then $\tilde{\mathcal{R}}^j = \mathcal{R}^j$*.*

Proof. Take $R \in \tilde{\mathcal{R}}^j$. We show that $\tilde{\mathcal{R}}^j$ is the image of $T\mathcal{R}^j_R$ through the map

$$T\mathcal{R}^j_R \to \mathrm{SO}(3)\,, \quad RW \mapsto Re^W\,,$$

in a neighborhood of R. In particular, this proves that $\tilde{\mathcal{R}}^j$ is an embedded submanifold of $\mathcal{R}^j$ and that the tangent spaces coincide, concluding the proof.

There exists a sequence $(R_h) \subset \mathrm{SO}(3)$ such that $R_h \in \mathcal{R}^j_h$ for every h and $R_h \to R$. For $h \ll 1$, take an orthonormal basis $\{R_h W^1_h, \dots, R_h W^m_h\}$ of $T\mathcal{R}^j_{h\,R_h}$, where $m = \dim \mathcal{R}^j_h = \dim \mathcal{R}^j$. Then $R_h W^i_h \to RW^i$ and since

$$0 = \frac{1}{h^{\gamma+1}} F_h(R(W^i_h)^2 U_j) \to F(R(W^i)^2 U_j)\,,$$

$\{RW^1, \dots, RW^m\}$ is an orthonormal basis of $T\mathcal{R}^j_R$. Now pick $W \in T\mathcal{R}^j_R$. By the convergence of the base we can construct a sequence (W_h) such that $W_h \to W$ and $R_h W_h \in T\mathcal{R}^j_h$ for every $h \ll 1$. By [21, Lemma 4.4] we have $R_h e^{W_h} \in \mathcal{R}^j_h$. Thus, passing to the limit, we get by Lemma 5.2 that $Re^W \in \mathcal{R}^j$, that is, by definition $Re^W \in \tilde{\mathcal{R}}^j$. ∎

The above results guarantee the well-posedness of the projection.

Proposition 5.6. *Let $(R_h) \subset \mathrm{SO}(3)$ be a sequence such that $R_h \to R$ and $R \in \mathcal{R}^j$ for some $j \in \{1, \dots, l\}$. Then $P^j_h(R_h)$ is well-defined for $h \ll 1$.*

Proof. It is sufficient to prove that $\mathrm{dist}_{\mathrm{SO}(3)}(R_h, \mathcal{R}^j_h) \to 0$. By Proposition 5.5, there exists a sequence of rotations $\tilde{R}_h$ such that $\tilde{R}_h \in \mathcal{R}^j_h$ for every h and $\tilde{R}_h \to R$. Since

$$\mathrm{dist}_{\mathrm{SO}(3)}(Q', Q) = |Q' - Q| + O(|Q' - Q|^2)$$

for every $Q, Q' \in \mathrm{SO}(3)$, we have

$$\mathrm{dist}_{\mathrm{SO}(3)}(R_h, \mathcal{R}^j_h) \le \mathrm{dist}_{SO(3)}(R_h, \tilde{R}_h) = |R_h - \tilde{R}_h| + O(|R_h - \tilde{R}_h|^2) \to 0\,.$$

∎

We are now ready to conclude the proof of Theorem 1.7.

Proof of Theorem 1.7. By Lemma 5.1, we have $E^\alpha_h(y_h) \le C$.

We prove that $\bar{R}^T U_j \in \mathcal{M}$. This will also show that $j \in \Lambda$. Indeed, let $RU_k \in K$ and define

$$y'_h(x', x_3) = RU_k \begin{pmatrix} x' \\ hx_3 \end{pmatrix}.$$

We get

$$\begin{aligned} J^\alpha_h(y_h) - \inf_y J^\alpha_h(y) &\ge J^\alpha_h(y_h) - J^\alpha_h(y'_h) \\ &\ge -\frac{1}{h^\alpha}\int_\Omega f_h \cdot y_h\,dx + \frac{1}{h^\alpha}\int_\Omega f_h \cdot RU_k \begin{pmatrix} x' \\ 0 \end{pmatrix} dx \\ &= -\frac{1}{h^\alpha}\int_\Omega f_h \cdot \bar{R}^T_h U_j^{-1} \begin{pmatrix} \max\{h^\gamma, h^{2\gamma-2}\} u_h \\ h^{\gamma-1} v_h \end{pmatrix} dx \\ &\quad + \frac{1}{h^\alpha}\int_\Omega f_h \cdot (RU_k - \bar{R}^T_h U_j) \begin{pmatrix} x' \\ 0 \end{pmatrix} dx\,. \end{aligned} \tag{42}$$

Multiplying by $h^{\gamma-1}$ and passing to the limit we deduce that

$$\int_\Omega f \cdot (RU_k - \bar{R}^T U_j) \begin{pmatrix} x' \\ 0 \end{pmatrix} dx \le 0\,,$$

where we have used convergence (8). Since $RU_k \in K$ is arbitrary, we have $\bar{R}^T U_j \in \mathcal{M}$.

By Proposition 5.6, the projection $P_h^j(R_h^T)$ is well-defined for $h \ll 1$. Let $W_h \in \mathbb{R}^{3\times 3}_{\text{skew}}$ with $|W_h| = 1$ be such that

$$P_h^j(\bar{R}_h^T)W_h \in N\mathcal{R}^j_{hP_h^j(\bar{R}_h^T)}$$

and $\bar{R}_h^T = P_h^j(\bar{R}_h^T)e^{d_h W_h}$, where $d_h = \text{dist}_{\text{SO}(3)}(\bar{R}_h^T, P_h^j(\bar{R}_h^T))$. Then, by (42) with $RU_k = P_h^j(\bar{R}_h^T)U_j$ and the fact that $F_h(P_h^j(\bar{R}_h^T)W_h U_j) = 0$ we have, expanding the exponential map,

$$\begin{aligned} J_h^\alpha(y_h) - \inf_y J_h^\alpha(y) &\geq J_h^\alpha(y_h) - J_h^\alpha\left(P_h^j(\bar{R}_h^T)U_j\begin{pmatrix} x' \\ hx_3\end{pmatrix}\right) \\ &\geq -C - \frac{d_h^2}{h^\alpha}\int_\Omega f_h \cdot P_h^j(\bar{R}_h^T)W_h^2 U_j\,dx + O\left(\frac{d_h^3}{h^{\gamma-1}}\right). \end{aligned} \tag{43}$$

Clearly $P_h^j(\bar{R}_h^T) \to \bar{R}^T$. Moreover, up to subsequences, $W_h \to W$ and by Lemma 5.3, $\bar{R}^T W \in N\mathcal{R}^j_{\bar{R}^T}$, thus $F(\bar{R}^T W^2 U_j) < 0$. We deduce by (43) that $d_h^2 = O(h^{\gamma-1})$. In particular, there exists $\beta \geq 0$ such that $h^{1-\gamma}d_h^2 \to \beta^2$ and so

$$h^{\frac{1}{2}(1-\gamma)}(\bar{R}_h^T - P_h^j(\bar{R}_h^T)) = \frac{d_h}{h^{\frac{1}{2}(\gamma-1)}}P_h^j(\bar{R}_h^T)W_h + O\left(\frac{d_h^2}{h^{\frac{1}{2}(\gamma-1)}}\right) \to \beta\bar{R}^T W\,.$$

We are left to prove the minimality property. We show it for $2 < \alpha < 4$. The other cases can be proved analogously. Firstly, note that

$$\frac{1}{h^\alpha}\int_\Omega f_h \cdot y_h\,dx - \frac{1}{h^\alpha}F_h(P_h^j(\bar{R}_h^T)U_j) \to \int_S f \cdot \bar{R}^T U_j^{-1} e_3 v\,dx - F(\beta^2 \bar{R}^T W^2 U_j)\,.$$

Indeed, we have the equality

$$\begin{aligned} \frac{1}{h^\alpha}\int_\Omega f_h \cdot y_h\,dx - F(P_h^j(\bar{R}_h^T)U_j) &= \frac{1}{h^\alpha}\int_\Omega f_h \cdot \bar{R}_h^T U_j^{-1}\begin{pmatrix} h^{2\gamma-2}u_h \\ h^{\gamma-1}v_h\end{pmatrix}dx \\ &\quad - \frac{1}{h^\alpha}\int_\Omega f_h \cdot (P_h^j(\bar{R}_h^T) - \bar{R}_h^T)U_j\begin{pmatrix} x' \\ 0\end{pmatrix}dx\,. \end{aligned}$$

The first term behaves as follows:

$$\begin{aligned} \frac{1}{h^\alpha}\int_\Omega f_h \cdot \bar{R}_h^T U_j^{-1}\begin{pmatrix} h^{2\gamma-2}u_h \\ h^{\gamma-1}v_h\end{pmatrix}dx &= \int_\Omega \frac{1}{h^{\gamma+1}} f_h \cdot \bar{R}_h^T U_j^{-1}\begin{pmatrix} h^{\gamma-1}u_h \\ v_h\end{pmatrix}dx \\ &\to \int_S f \cdot \bar{R}^T U_j^{-1} e_3 v\,dx\,. \end{aligned}$$

On the other hand, since $F_h(P_h^j(\bar{R}_h^T)W_h U_j) = 0$, we have

$$\frac{1}{h^\alpha}F_h(P_h^j(\bar{R}_h^T)U_j) = \frac{d_h^2}{h^{\gamma-1}}\int_\Omega \frac{1}{h^{\gamma+1}} f_h \cdot P_h^j(\bar{R}_h^T)W_h^2 U_j\begin{pmatrix} x' \\ 0\end{pmatrix}dx + O\left(\frac{d_h^3}{h^{\gamma-1}}\right),$$

that converges to $F(\beta^2 \bar{R}^T W^2 U_j)$. Thus, by Theorem 1.3–(ii) we have

$$\liminf_{h\to 0}(J_h^\alpha(y_h) + F(P_h^j(\bar{R}_h^T)U_j)) \geq J_j^{CVK}(v, \bar{R}^T, \beta W)\,. \tag{44}$$

Take an admissible quadruplet (i, v', R', W') and let y_h' be the recovery sequence for v' provided by Theorem 1.3–(iii). By Proposition 5.5, we can construct a sequence of rotations R_h' converging to R' such that $R_h' \in \mathcal{R}_h^i$. Note that, by (W3),

$$E_h^\alpha(y_h') = E_h^\alpha(R_h' y_h')\,,$$